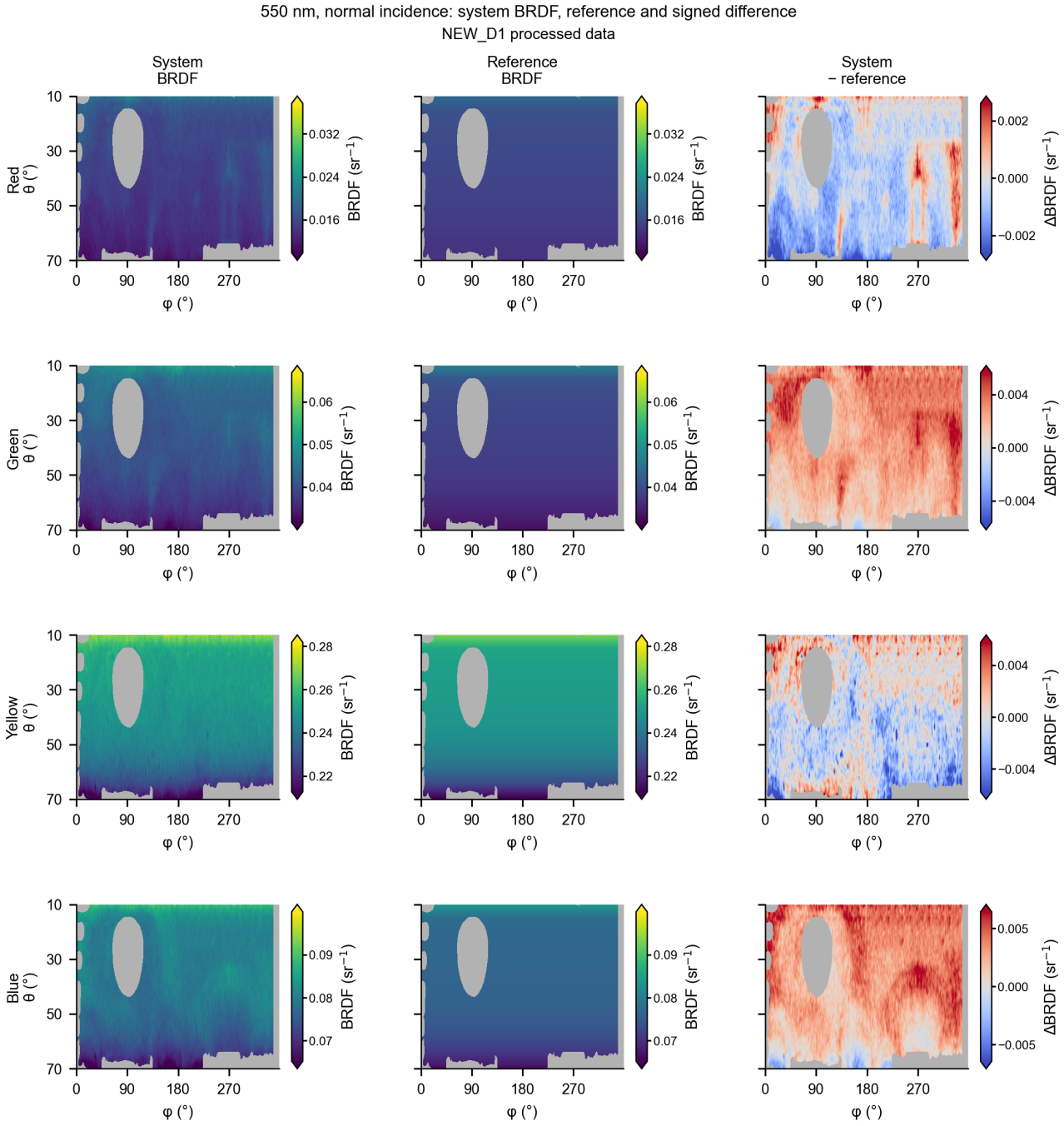
Supplementary Material



Supplementary Fig. S1. Normal-incidence BRDF distributions for the red, green, yellow and blue tiles at 550 nm. System BRDF is shown after denoising and other data processing (NEW_D1 processed data). Rows correspond to samples; columns show system BRDF, the rotationally reconstructed reference and the signed system-minus-reference difference. BRDF and difference values are in sr^{-1}. Paired system and reference maps share a color scale within each row; difference scales are symmetric about zero. Gray cells are outside the common display comparison mask. The reference azimuthal distribution is reconstructed from principal-plane scans under the normal-incidence azimuthal-symmetry approximation.

Supplementary Table S1

Common-domain integrated BSDF results for the validation samples across eight spectral channels.

Data series NEW_D1 processed data. Normal incidence; four BRDF tiles and four BTDF PMMA samples. All integrals are dimensionless.

Sample	BSDF	Wavelength (nm)	System $Q_{CD,\,sys}$	Reference $Q_{CD,\,ref}$	Signed relative discrepancy (%)	Low signal
Red tile	BRDF	430	0.033312421	0.036757436	-9.3723	No
Red tile	BRDF	460	0.027496987	0.028138236	-2.2789	No
Red tile	BRDF	490	0.030426474	0.02922143	+4.1238	No
Red tile	BRDF	520	0.036062008	0.029974522	+20.3089	No
Red tile	BRDF	550	0.034647852	0.03566702	-2.8575	No
Red tile	BRDF	590	0.12990241	0.11762443	+10.4383	No
Red tile	BRDF	630	0.54355099	0.52940965	+2.6712	No
Red tile	BRDF	660	0.58095615	0.57868792	+0.3920	No
Green tile	BRDF	430	0.039271457	0.049587219	-20.8033	No
Green tile	BRDF	460	0.025971821	0.027642418	-6.0436	No
Green tile	BRDF	490	0.034176345	0.033016934	+3.5116	No
Green tile	BRDF	520	0.065848377	0.074557662	-11.6813	No
Green tile	BRDF	550	0.093529875	0.087640572	+6.7198	No
Green tile	BRDF	590	0.039449105	0.046474563	-15.1168	No
Green tile	BRDF	630	0.04406058	0.046643161	-5.5369	No
Green tile	BRDF	660	0.065712782	0.06914104	-4.9584	No
Yellow tile	BRDF	430	0.084576205	0.088403781	-4.3297	No
Yellow tile	BRDF	460	0.12035359	0.11733655	+2.5713	No
Yellow tile	BRDF	490	0.26449043	0.21761572	+21.5401	No
Yellow tile	BRDF	520	0.43538969	0.43605651	-0.1529	No
Yellow tile	BRDF	550	0.5621886	0.56358128	-0.2471	No
Yellow tile	BRDF	590	0.58515678	0.58397556	+0.2023	No
Yellow tile	BRDF	630	0.59632737	0.59753225	-0.2016	No
Yellow tile	BRDF	660	0.59643184	0.59998679	-0.5925	No
Blue tile	BRDF	430	0.34828502	0.35752707	-2.5850	No
Blue tile	BRDF	460	0.36807311	0.37004666	-0.5333	No
Blue tile	BRDF	490	0.32074253	0.33653682	-4.6932	No
Blue tile	BRDF	520	0.26146031	0.25711416	+1.6904	No
Blue tile	BRDF	550	0.17976846	0.17332443	+3.7179	No
Blue tile	BRDF	590	0.11255422	0.12108892	-7.0483	No
Blue tile	BRDF	630	0.092789694	0.093525234	-0.7865	No
Blue tile	BRDF	660	0.098998242	0.10028924	-1.2873	No
PMMA 4	BTDF	430	0.37822705	0.3687774	+2.5624	No
PMMA 4	BTDF	460	0.5052958	0.45097991	+12.0440	No
PMMA 4	BTDF	490	0.52012377	0.47594946	+9.2813	No
PMMA 4	BTDF	520	0.5124331	0.47984769	+6.7908	No
PMMA 4	BTDF	550	0.43687344	0.42081081	+3.8171	No
PMMA 4	BTDF	590	0.30841192	0.33415746	-7.7046	No
PMMA 4	BTDF	630	0.28327135	0.29670483	-4.5276	No

Sample	BSDF	Wavelength (nm)	System $Q_{CD,\,sys}$	Reference $Q_{CD,\,ref}$	Signed relative discrepancy (%)	Low signal
PMMA 4	BTDF	660	0.31528455	0.3251162	-3.0240	No
PMMA 5	BTDF	430	0.040773316	0.049663798	-17.9013	Yes
PMMA 5	BTDF	460	0.063787655	0.065057964	-1.9526	No
PMMA 5	BTDF	490	0.1209878	0.11274108	+7.3147	No
PMMA 5	BTDF	520	0.2699544	0.28403765	-4.9582	No
PMMA 5	BTDF	550	0.44789856	0.413345	+8.3595	No
PMMA 5	BTDF	590	0.35014721	0.3632542	-3.6082	No
PMMA 5	BTDF	630	0.26568301	0.277478	-4.2508	No
PMMA 5	BTDF	660	0.27594889	0.28840276	-4.3182	No
PMMA 6	BTDF	430	0.18700823	0.19423999	-3.7231	No
PMMA 6	BTDF	460	0.20901065	0.20840196	+0.2921	No
PMMA 6	BTDF	490	0.24128977	0.23323609	+3.4530	No
PMMA 6	BTDF	520	0.33345355	0.33972456	-1.8459	No
PMMA 6	BTDF	550	0.49661351	0.45867786	+8.2707	No
PMMA 6	BTDF	590	0.52936882	0.48273022	+9.6614	No
PMMA 6	BTDF	630	0.54254112	0.49865103	+8.8018	No
PMMA 6	BTDF	660	0.54830907	0.50547124	+8.4748	No
PMMA 7	BTDF	430	0.002248884	0.013632853	-83.5039	Yes
PMMA 7	BTDF	460	0.00025222042	0.0073928493	-96.5883	Yes
PMMA 7	BTDF	490	0.0031441119	0.0056974492	-44.8154	Yes
PMMA 7	BTDF	520	0.0036449233	0.0038186628	-4.5497	Yes
PMMA 7	BTDF	550	-0.00035196306	0.0031520405	-111.1662	Yes
PMMA 7	BTDF	590	0.024953908	0.02816128	-11.3893	No
PMMA 7	BTDF	630	0.31742006	0.29314764	+8.2799	No
PMMA 7	BTDF	660	0.39235217	0.36544367	+7.3632	No

Table notes

The integration uses the same sample–channel validity mask for the system and reference within $10° \le \theta \le 70°$, with $\cos(\theta)\, d\Omega$ weights. No normalization to a full hemisphere is applied. The reference azimuthal distribution is reconstructed from principal-plane scans under the normal-incidence rotational-symmetry approximation for diffuse samples.

The signed relative discrepancy in each row is $100 \times (Q_{CD,\,sys} / Q_{CD,\,ref} - 1)$. Positive values indicate a larger system integral. The source permits this ratio only when $|Q_{CD,\,ref}| > 10^{-12}$; all 64 rows meet that criterion. Signed system integrals, including negative low-signal estimates, are retained.

The signal-weighted absolute discrepancy for any group is $E = 100 \times \Sigma|Q_{CD,\,sys} - Q_{CD,\,ref}| / \Sigma|Q_{CD,\,ref}|$. It is not the arithmetic mean of the signed row percentages. All cases, including the low-signal flags, enter these sums. BRDF: 2.81% (32 cases); BTDF: 6.74% (32 cases); aggregate: 5.00% (64 cases).

“Yes” marks the six prior low-signal cases. Cells below the earlier 20.5 DN criterion account for at least 50% of the original-domain $\cos(\theta)\, d\Omega$ weight. All flagged cases enter the aggregate calculations.

The system common-domain integrals are reported after denoising and other data processing (NEW_D1 processed data), alongside the reference integrals. Full-precision values, low-signal flags and source hashes accompany the table in CSV and README files.